

EE4410 CYBER SECURITY OF POWER GRIDS

Assignment 4: Operational Technology Traffic Analysis

June 11, 2025

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[Link to Google Colab Notebook](#)

This lab report focuses on the analysis of operational technology (OT) network traffic, specifically GOOSE messages, using time-series analysis and graph-based models. The aim is to identify unusual patterns and better understand how the network functions.

1 Network Traffic Time-Series Analysis

This section analyzes IT and OT traffic characteristics using packet length analysis and time-series resampling.

1.1 Traffic Data Visualization

The boxplot in [Figure 1](#) displays the packet length distributions across various protocols observed in the dataset. TLSv1.2 and HTTP show the highest packet lengths and widest spreads, with TLSv1.2 reaching values beyond 1700 bytes and many outliers. HTTP also shows high and relatively stable packet sizes.

In contrast, GOOSE traffic appears tightly clustered around a specific length (~180 bytes), with minimal spread and a single outlier. The SMB2 protocol also stands out. Its packet sizes are considerably higher than GOOSE and most other protocols, and its distribution exhibits a wider interquartile range, spanning roughly from 150 to 450 bytes, with several outliers reaching into the thousands.

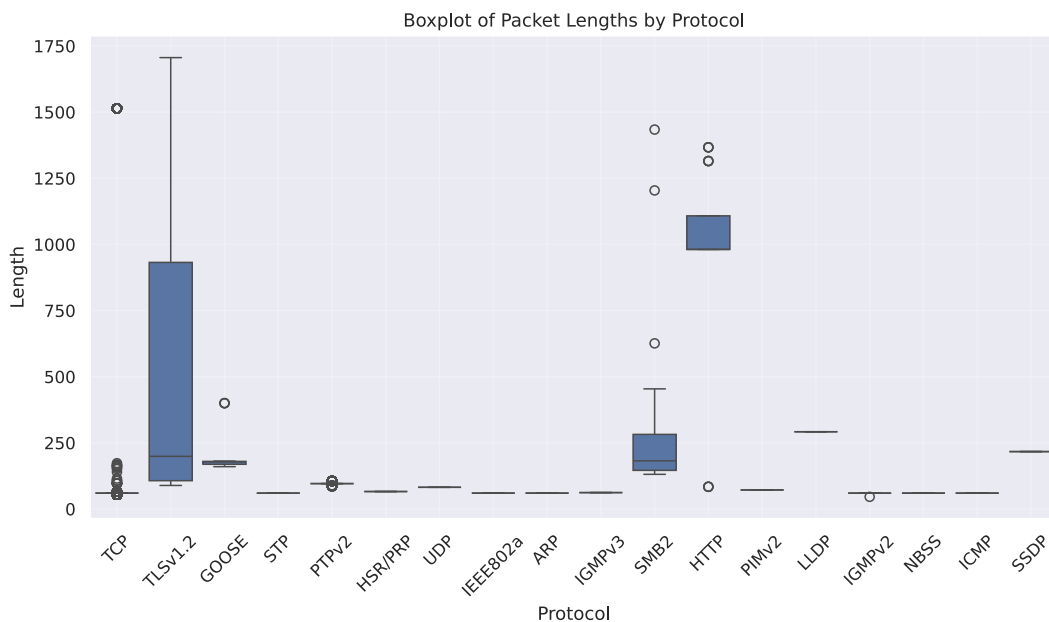


Figure 1: Boxplot of packet lengths across protocols

The time-series plot in [Figure 2](#) shows how GOOSE traffic changes over time. A repeating, burst-like pattern was clearly visible, indicating cyclic transmission.

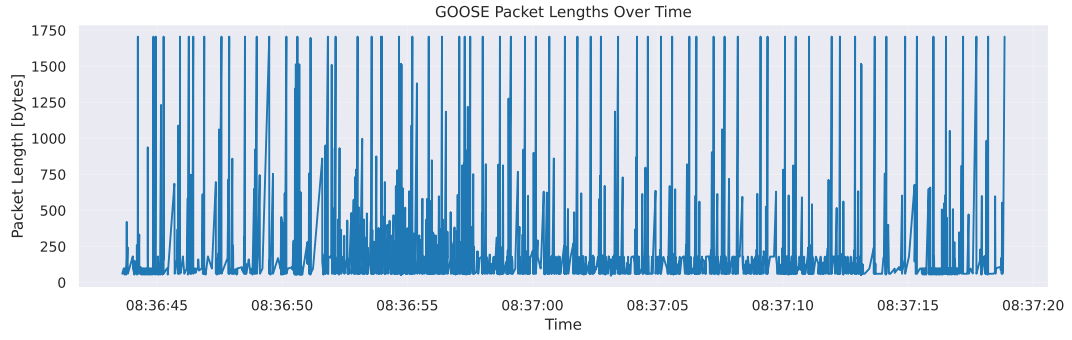


Figure 2: Time-series plot of GOOSE packet lengths

Repetitive structures are typical of status updates or synchronization messages. These are key features of GOOSE communication in power systems.

1.2 Resampling of The Times-Series Data

To assess traffic behavior across different time granularities, the packet length data was resampled using five intervals: 100ms, 200ms, 500ms, 1000ms, and 2000ms. The total packet length was aggregated over each window to observe temporal characteristics in both GOOSE and normal traffic. As shown in [Figure 3](#), finer resampling intervals (e.g., 100ms and 200ms) preserved the bursty nature and variability in the traffic. In contrast, coarser intervals (e.g., 1000ms and 2000ms) tended to smooth out short-term variations, which can obscure meaningful anomalies or cycles.

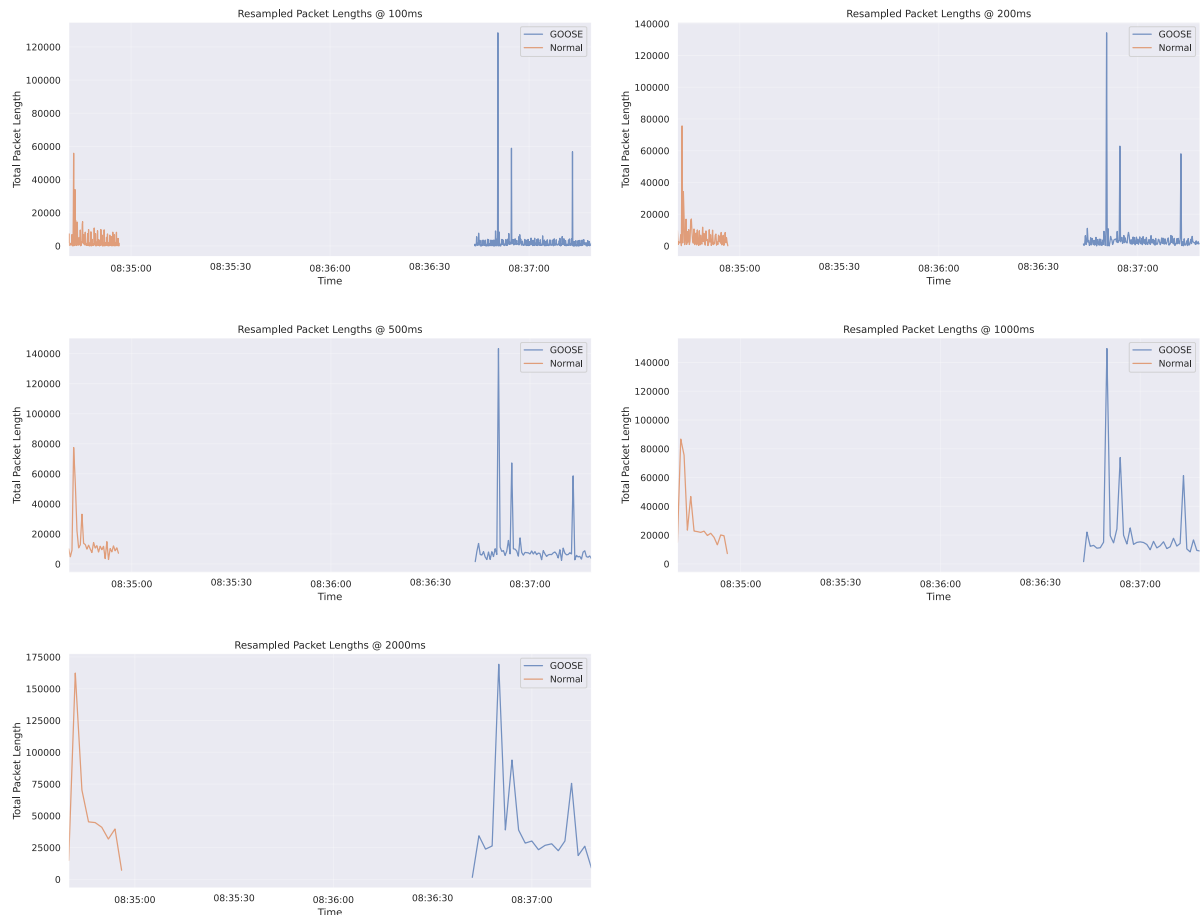


Figure 3: Comparison of resampled packet lengths at various intervals for both GOOSE and normal traffic

As the interval increases, the fine-grained fluctuations are smoothed out. Smaller intervals like 100 ms and 200 ms preserve temporal resolution and reveal micro-bursts, while larger intervals cause information loss due to aggregation.

Among the tested intervals, 200 ms was identified as a suitable compromise. It preserves temporal detail without excessive noise and retains distinguishable behavior between protocols. The plot in [Figure 4](#) shows protocol-specific traffic dynamics at this resolution, revealing bursty behavior in GOOSE and relatively constant traffic in others (e.g., TCP, HTTP).

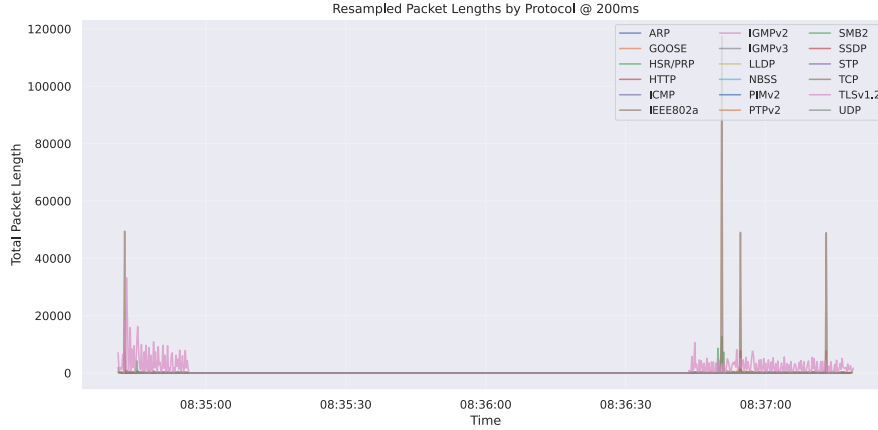


Figure 4: Resampled traffic for all protocols at 200 ms

For GOOSE traffic in isolation, the 200 ms interval also proved effective. As depicted in [Figure 5](#), periodic bursts and anomalies are retained with sufficient resolution, while noise is minimized. This confirms the selection of 200 ms as an optimal resampling setting for further GOOSE-specific analysis.

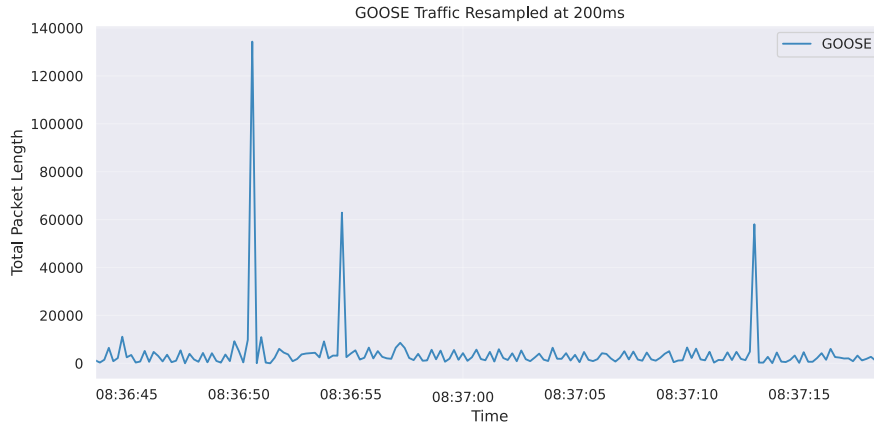


Figure 5: GOOSE traffic resampled at 200 ms

1.3 Comparison and Anomaly Detection

Based on the boxplots and resampled visualizations (see [Figure 1](#) and [Figure 4](#)), a clear distinction between OT and IT protocol behavior can be observed. **OT protocols** such as GOOSE exhibit highly cyclic and bursty traffic patterns. As shown in [Figure 5](#), the GOOSE signal is periodic, with consistently sized packets appearing at regular intervals. This behavior aligns with the deterministic nature of substation automation systems where time-critical messages are broadcast repeatedly.

In contrast, **IT protocols** such as HTTP, TCP, and SMB demonstrate a broader spread in packet length and more irregular fluctuations. As visible in [Figure 1](#), the interquartile ranges for these protocols are

wide, and several extreme outliers are present. These characteristics reflect the stochastic and user-driven nature of IT communication, typical in office or administrative environments.

As for anomaly detection, a comparison of the traffic in the `goose.csv` and `normal.csv` datasets reveals specific differences. The normal traffic shows consistent GOOSE activity with minor variation, indicating stable conditions. On the other hand, the `goose` traffic includes sharp spikes in packet aggregation that deviate from the expected pattern (see [Figure 3](#) and [Figure 5](#)). These abrupt changes suggest abnormal behavior, possibly due to cyber attacks such as spoofing or replay. Therefore, the `goose.csv` file is most likely to contain anomalous traffic.

2 Traffic Dispersion Graph

A traffic flow graph was generated from the dataset using the NetworkX library. Each node represents a device identified by its source or destination address, while directed edges represent packet transfers between these devices.

2.1 Initial Flow Graph

In the first version of the graph, raw communication flows between GOOSE devices were visualized using NetworkX. As shown in [Figure 6](#), the network consisted of 9 nodes and 5 directed edges, indicating direct device-to-device transmissions among a relatively sparse set of hosts.

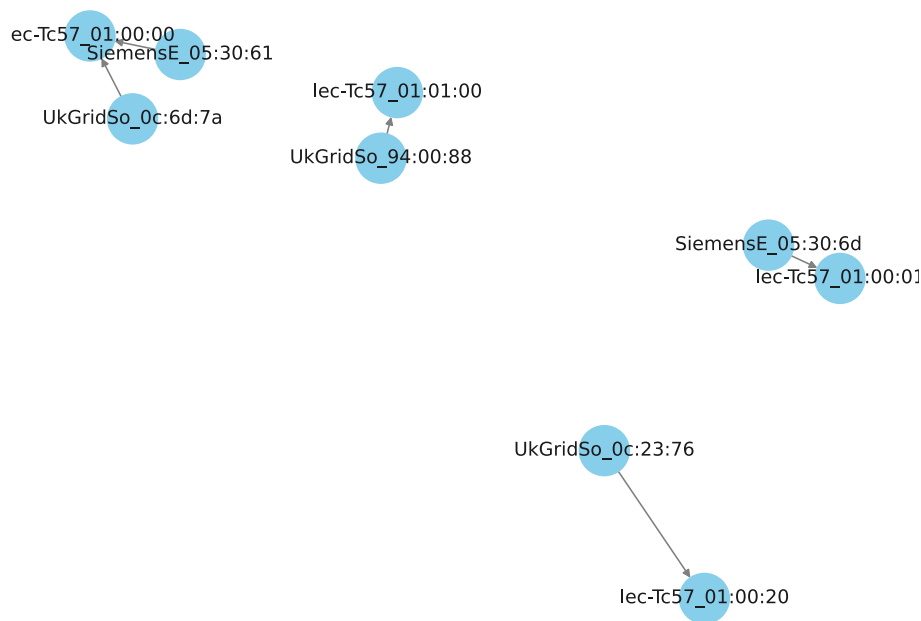


Figure 6: Initial GOOSE traffic flow graph based on original communication pairs

2.2 Switch Node Graph

To better reflect typical OT network topologies, a switch node was added and all traffic was routed through it. This resulted in a denser graph with 10 nodes and 9 edges, as seen in [Figure 7](#). The switch node centralized all communication, significantly increasing its connectivity compared to other nodes.

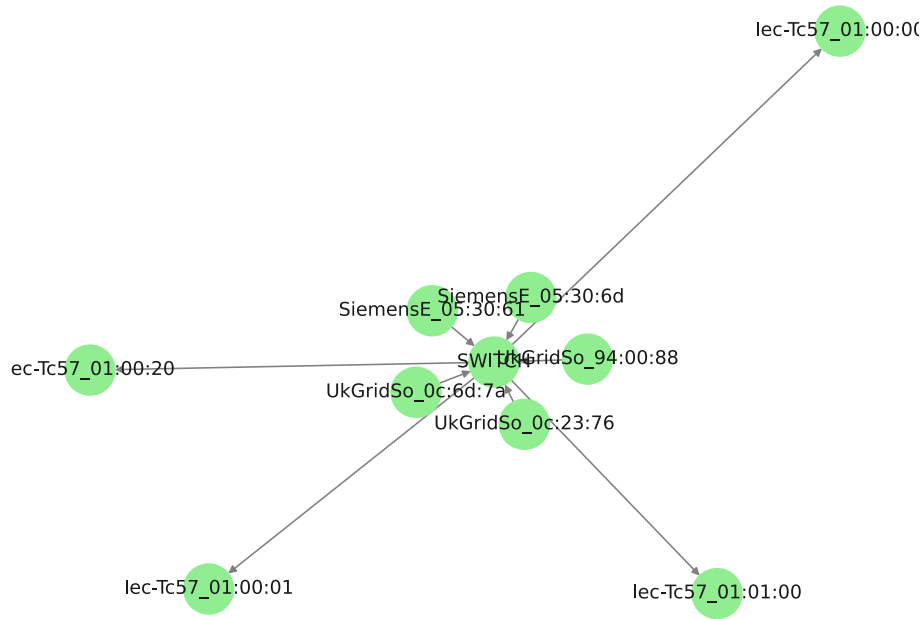


Figure 7: GOOSE traffic graph with all flows routed through a switch node

2.3 Nodes and Characteristics

The two graphs show different communication patterns. In the raw graph (Figure 6), the traffic was sparse and devices communicated directly with each other. In the switch-based graph (Figure 7), all traffic passed through a central switch, forming a star-shaped structure. End devices acted as either senders or receivers, while the switch connected them all. This setup matches typical GOOSE communication in substations, where messages are shared through a central switch using multicast.